# Spandan **Senapati**

CS Ph.D. student, University of Southern California

*Los Angeles, California, USA*

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## Education

**University of Southern California** — *Los Angeles, California*

Ph.D. in Computer Science, advised by Prof. Haipeng Luo and Prof. Vatsal Sharan — *Aug 2023 - Present*

- Research Interests: Statistical Learning Theory, Online Learning, Trustworthy Machine Learning

**Indian Institute of Technology Kanpur** — *Kanpur, Uttar Pradesh*

Bachelor of Technology in Computer Science and Engineering — *July 2018 - May 2022*

- Cumulative Performance Index (CPI): **9.1/10.0** (graduated with **Distinction**)
- Awarded **Proficiency Medal** at the 55th Convocation for the best undergraduate project done by a graduating student in the CSE department
- **Academic Excellence Award** (2018-19, 2019-20): Awarded for excellent academic performance in an academic year

## Experience

**Apple** — *Cupertino, CA*

Internship, Mentors: Parikshit Gopalan, Satyen Kale — *May 2026 - Aug 2026*

**Indian Institute of Technology Kanpur** — *SPiN Lab, IIT Kanpur*

Graduate Research Assistant, Advisor: Prof. Ketan Rajawat — *Aug 2022 - May 2023*

**Tata Institute of Fundamental Research** — *Remote*

Internship, Advisor: Prof. Rahul Vaze — *Aug 2022 - May 2023*

**Microsoft** — *Remote*

Data Scientist intern, Related Search Team — *May 2021 - July 2021*

## Publications and Preprints

**Efficient Swap Multicalibration of Elicitable Properties** — *[Arxiv]*

Lunjia Hu$^{\alpha-\beta}$, Haipeng Luo$^{\alpha-\beta}$, **Spandan Senapati**$^{\alpha-\beta}$, Vatsal Sharan$^{\alpha-\beta}$ — *Nov 2025*

Annual Conference on Learning Theory (COLT), 2026

**Improved Bounds for Swap Multicalibration and Swap Omniprediction** — *[Arxiv]*

Haipeng Luo$^{\alpha-\beta}$, **Spandan Senapati**$^{\alpha-\beta}$, Vatsal Sharan$^{\alpha-\beta}$ — *May 2025*

Annual Conference on Neural Information Processing Systems (NeurIPS), 2025 (**Spotlight presentation**)

**Simultaneous Swap Regret Minimization via KL-Calibration** — *[Arxiv]*

Haipeng Luo$^{\alpha-\beta}$, **Spandan Senapati**$^{\alpha-\beta}$, Vatsal Sharan$^{\alpha-\beta}$ — *Feb 2025*

Annual Conference on Neural Information Processing Systems (NeurIPS), 2025 (**Spotlight presentation**)

**Optimal Multiclass U-Calibration Error and Beyond** — *[Arxiv]*

Haipeng Luo$^{\alpha-\beta}$, **Spandan Senapati**$^{\alpha-\beta}$, Vatsal Sharan$^{\alpha-\beta}$ — *Sep 2024*

Annual Conference on Neural Information Processing Systems (NeurIPS), 2024

**Sharpened Lazy Incremental Quasi-Newton Method** — *[PMLR][Arxiv]*

Aakash Lahoti*, **Spandan Senapati***, Ketan Rajawat, Alec Koppel — *Jan 2024*

International Conference on Artificial Intelligence and Statistics (AISTATS)

**Online Convex Optimization with Switching Cost and Delayed Gradients** — *[PER][PEVA][Arxiv]*

**Spandan Senapati**, Rahul Vaze — *Aug 2023*

IFIP Performance 2023. Full paper: Elsevier Performance Evaluation (PEVA), Extended abstract: ACM SIGMETRICS Performance Evaluation Review (PER)

**Proximal Algorithms for Smoothed Online Convex Optimization with Predictions** — *[IEEE Xplore][Arxiv]*

**Spandan Senapati**, Ashwin Shenai, Ketan Rajawat — *Aug 2023*

IEEE Transactions on Signal Processing

Note: $\alpha - \beta$ denotes alphabetical ordering, which is custom in theoretical computer science research; $*$ denotes equal contribution

## Honors & Awards

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|---|---|---|
| 2022 | **Proficiency Medal**, Best undergraduate project done by a graduating student from the CSE department | *IIT Kanpur, India* |
| 2018, 19 | **Academic Excellence Award**, For excellent academic performance in an academic year | *IIT Kanpur, India* |
| 2018 | **All India Rank 161**, Joint Entrance Examination Main, among 1.5 Million Candidates | *India* |
| 2018 | **All India Rank 191**, Joint Entrance Examination Advanced, among 200000 Candidates | *India* |
| 2017 | **All India Rank 24**, KVPY Scholarship, Indian Institute of Science and Government of India | *Bangalore, India* |
| 2016 | **NTSE Scholar**, Awarded to the top 1000 students in the nation by the Government of India | *India* |
| 2016 | **IJSO OCSC**, Selected for the team selection camp of the International Junior Science Olympiad(top 35 India) | *Mumbai, India* |
| 2017, 18 | **National Top 1% ,** National Standard Examination in Physics(NSEP) among 200000 Candidates | *India* |
| 2018 | **National Top 1% ,** National Standard Examination in Chemistry(NSEC) among 200000 Candidates | *India* |
| 2017, 18 | **State Top 1% ,** National Standard Examination in Astronomy(NSEA) | *India* |
| 2017 | **3rd in State**, Represented Odisha State in the Indian National Maths Olympiad(INMO) | *India* |
| 2016 | **State Top 35**, Represented Odisha State in the Indian National Maths Olympiad (INMO) | *India* |

## Travel Grants

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| 2024 | **GSG Professional Development Fund**, Issued by USC |
| 2023 | **National Science Foundation (NSF) Student Travel Award**, Issued by IFIP Performance 2023 |

## Talks

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|---|---|---|
| 2026 | **Stanford Theory Lunch**, Efficient Swap Multicalibration of Elicitable Properties | *Palo Alto* |
| 2026 | **Apple Machine Learning Research**, Efficient Swap Multicalibration of Elicitable Properties | *Remote* |
| 2026 | **USC Theory Lunch**, Efficient Swap Multicalibration of Elicitable Properties | *Los Angeles* |
| 2024 | **USC Theory Lunch**, Optimal Multiclass U-Calibration Error and Beyond | *Los Angeles* |
| 2023 | **IFIP Performance 2023**, Online Convex Optimization with Switching Cost and Delayed Gradients | *Chicago* |

## Teaching

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| Spring'26 | **Teaching Assistant**, Machine Learning (CSCI 567) | *USC* |
| Fall'24 | **Teaching Assistant**, Theoretical Machine Learning (CSCI 678) | *USC* |

## Skills

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| **Programming** | C, C++, Python |
| **Machine Learning Framework** | Pytorch |

## Reviewing

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| **Conference** | NeurIPS 2024, 2025 (**Top Reviewer**), 2026 (Emergency Reviewer), ALT 2026 (Emergency Reviewer), FORC 2026 (External Reviewer), FOCS 2025 (External Reviewer), SODA 2026 (External Reviewer) |
| **Journal** | IEEE BITS the Information Theory Magazine |

## Service

### SHINE Mentor — *USC*

USC Summer High School Intensive in Next Generation Engineering (SHINE) — *June 2024 - July 2024*

- Mentored a high school student in building an online learning-based program that can beat a human user repeatedly in a game of rock-paper-scissors

### Coordinator — *IIT Kanpur*

Special Interest Group in Machine Learning (SIGML) — *May 2021 - Nov 2021*

- Responsible for the smooth conduction of talks in different active areas of research in Machine Learning

### Student Guide — *IIT Kanpur*

Counselling Service — *July. 2019 - July 2020*

- Responsible for the guidance and mentorship at a personal level of 5 freshman undergraduates and the smooth conduction of various events in the Orientation Session for the incoming Batch of Y19